DRIVER DROWSINESS DETECTION

Research Paper Comparison Table

S.no	Paper Title	Author(s)/Year	Journal Name	Methodology & Technology	Advantages	Limitations
1	Deep Learning-Based Driver Drowsiness Detection Using CNN	M. Thejovathi et al. (2025)	IEEE Access	CNN, facial landmarks, EAR	High accuracy; real-time	Poor in low light
2	Real-Time Driver Fatigue Detection Using Vision-Based Deep Learning	R. Kavitha et al. (2025)	Expert Systems with Applications	CNN, LSTM	Temporal learning	High computation
3	Driver Drowsiness Detection Using YOLO and CNN	P. Solmon et al. (2025)	Sensors	YOLO, CNN	Fast detection	Sunglasses affect accuracy
4	Vision-Based Driver Monitoring System	T. Zambrano et al. (2025)	Applied Sciences	CNN, landmarks	Non-invasive	Head pose sensitive
5	Hybrid Deep Learning Model	A. Bouyer et al. (2025)	IEEE T-ITS	CNN-LSTM	High accuracy	Complex model
6	Multi-Modal Driver Fatigue Detection	A. Sah et al. (2026)	Pattern Recognition	CNN, BiLSTM	Robust	Needs sensors
7	Transfer Learning for Alertness	S. Meenakshi et al. (2026)	Computers in Biology and Medicine	MobileNetV2	Lightweight	Depends on pretrained model
8	Facial Landmark Analysis	A. Karthik et al. (2026)	EAAI	MediaPipe, CNN	Efficient	Head movement issues
9	Attention-Based Framework	P. Vignesh et al. (2026)	Knowledge-Based Systems	Attention CNN-LSTM	Better features	Higher inference

10	Intelligent Drowsiness Detection	G. Oise et al. (2026)	IEEE Access	CNN, OpenCV	Easy deployment	Weather sensitive

RESEARCH GAP IDENTIFICATION:

Across all ten studies, no single system combines high accuracy, real-time edge deployment, and robustness together. Simpler CNN/OpenCV-based models [1] are lightweight but ignore temporal and multi-modal cues, while advanced hybrids (CNN-LSTM-ViT, ViT-CNN, EM-ViT) [2,6,7,8,10] boost accuracy but demand heavy computation unsuited for low-power in-vehicle hardware. Emotion-aware systems [4] still fail on subtle expressions like fear or sadness, and keyframe- or landmark-only approaches [5,9] degrade under varied lighting or cross-dataset conditions (e.g., NTHU-DDD). Multi-modal and explainable systems [7,8] add interpretability but require specialized IR sensors and increase system complexity. Large-scale pretrained models [10] need massive datasets(1.2M videos) and target-specific quantization, limiting general portability. This highlights the need for a generalizable, illumination-robust, and real-time model that unifies temporal, spatial, and emotional cues without sacrificing edge efficiency.

SDG MAPPING:

This driver drowsiness detection research primarily aligns with SDG 3 (Good Health and Well-Being), as early fatigue detection directly reduces road accident-related injuries and deaths. It strongly connects to SDG 9 (Industry, Innovation and Infrastructure), since the papers advance AI/deep learning innovation (CNN, ViT, LSTM, edge computing) for safer transportation infrastructure. SDG 11 (Sustainable Cities and Communities) is relevant through target 11.2, which calls for safe, accessible transport systems and reduced road traffic fatalities. The edge-AI and low-latency deployment focus (papers [8], [10]) also touches SDG 12 (Responsible Consumption and Production) via energy-efficient, optimized on-device inference (INT8 quantization, TensorRT) reducing computational waste.

SDGs used in this research:

SDG 3 – Good Health and Well-Being (accident/injury prevention)

SDG 9– Industry, Innovation and Infrastructure (AI-driven technological advancement)

SDG 11– Sustainable Cities and Communities (road safety, target 11.2)

SDG 12 – Responsible Consumption and Production (efficient edge computing)

REFERENCES:

1.M. Thejovathi, "Deep Learning-based Driver Drowsiness Detection and Real-time Alert System," 2026.

2.V. J. Pawar, "A Unified Deep Learning-Based Framework for Anticipatory Driver Drowsiness Detection Leveraging the Composite Drowsiness Score," 2026.

3.P. Solmon, "Driver Drowsiness Detection using Machine Learning and Deep Learning," 2026.

4.T. Zambrano, "Driver Monitoring System Using Computer Vision for Real-Time Detection of Fatigue, Distraction and Emotion via Facial Landmarks and Deep Learning," 2026.

5.A. Bouyer, "A Smart Vision System for Early Driver Drowsiness.Detection

using CNN–Transformer Hybrid Model with Attention-Based Learning," 2026.

6.D. Ganesh, "Vision Transformer-based Classification Approach for Driver Drowsiness Detection," 2026.

7.A. Sah, "An Approach for Identification of Driver Drowsiness, Emotion State, and Behavior Patterns Based on Vision Transformers, CNN and Multi-Modal Sensor Fusion using RAG-Based Interpretability," 2026.

8.D. Agarwal, "A Cross-Modal Edge Intelligence Framework for Driver Drowsiness Assessment using Transformer-based Vision and In-Cabin Sensing," 2026.

9.I. A. Haman, "Driver Drowsiness Detection Using Facial Temporal Features and Sequence

Based Deep Learning Models," 2026.

10.G. Oise, "Enhancing Road Safety through Deep Learning-Based Drowsiness Detection Using Vision AI," 2026.